

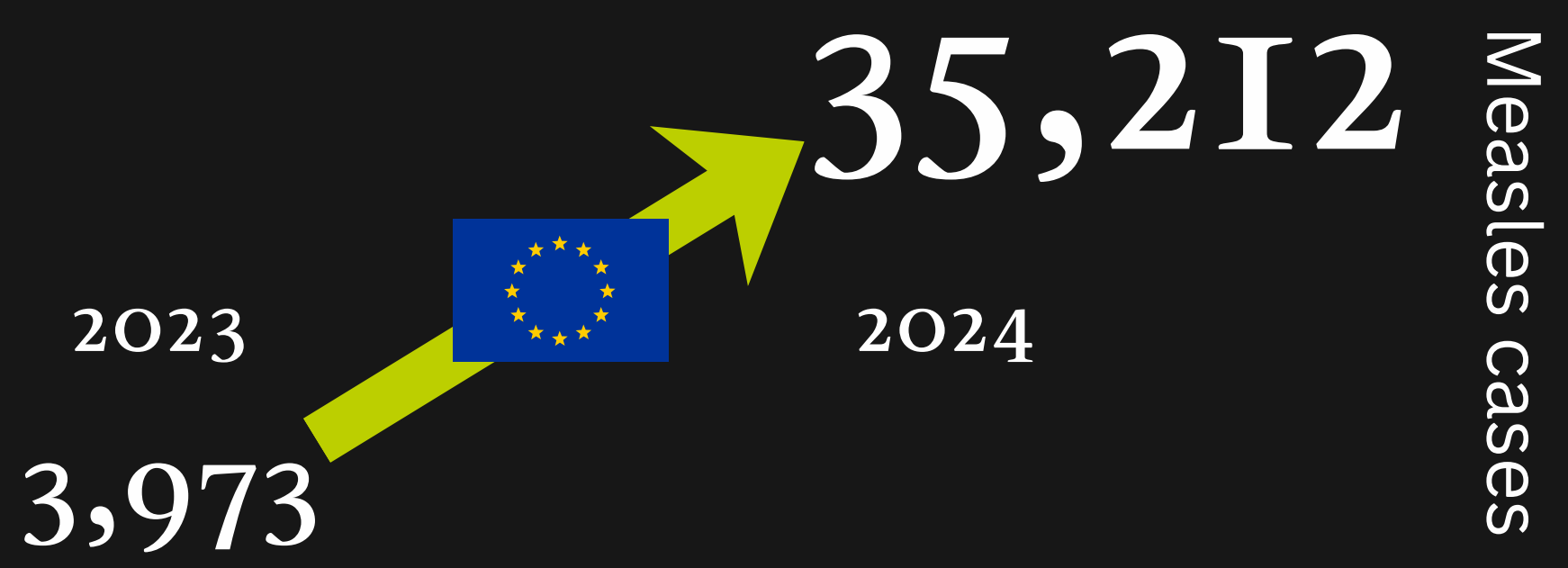
# A measles pseudovirus neutralization assay captures strain-specific neutralizing antibody activity with good repeatability.

## BACKGROUND:

- Measles: highly contagious, vaccine-preventable disease caused by the measles virus (MeV).
- Recent outbreaks worldwide, even in countries with high vaccine coverage like Belgium.
- Reports of infections in vaccinated individuals raise concerns about a possible primary vaccine failure, antigenic mismatch between vaccine and circulating strains, and/or waning immunity.

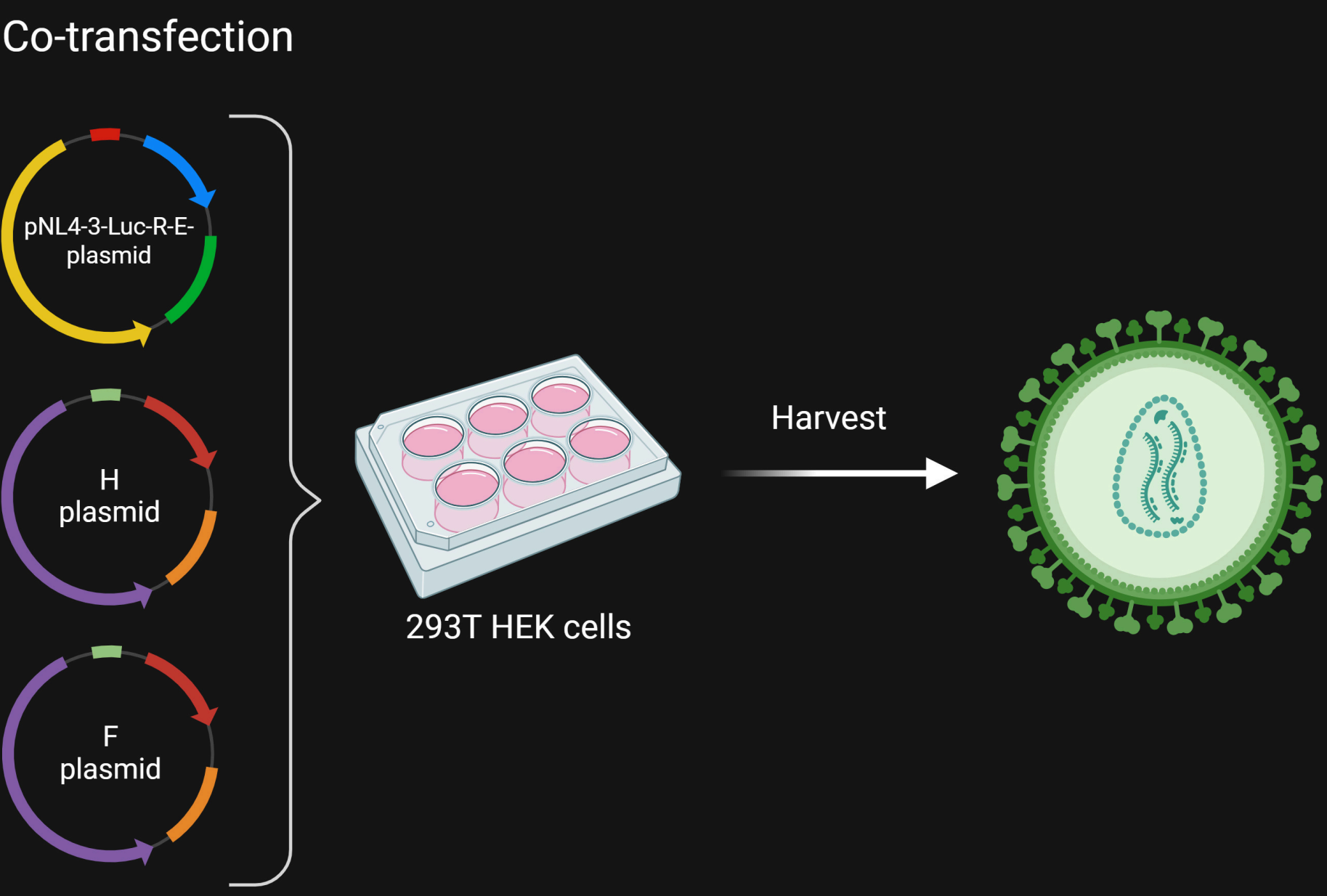
- Measuring functional neutralization against vaccine and circulating strains is essential.
- Plaque reduction neutralization testing (PRNT): WHO-accepted gold standard, but requires live MeV, is labor-intensive, NO high throughput capability.

- We developed a **pseudovirus (PV)-based neutralization assay** using **lentiviral particles** bearing **MeV Hemagglutinin (H)** and **Fusion (F)** surface glycoproteins from the **Edmonston vaccine strain** and the **D8 strain** (predominant in Belgium in 2024). Particles carry a **luciferase reporter**, enabling quantification of entry.



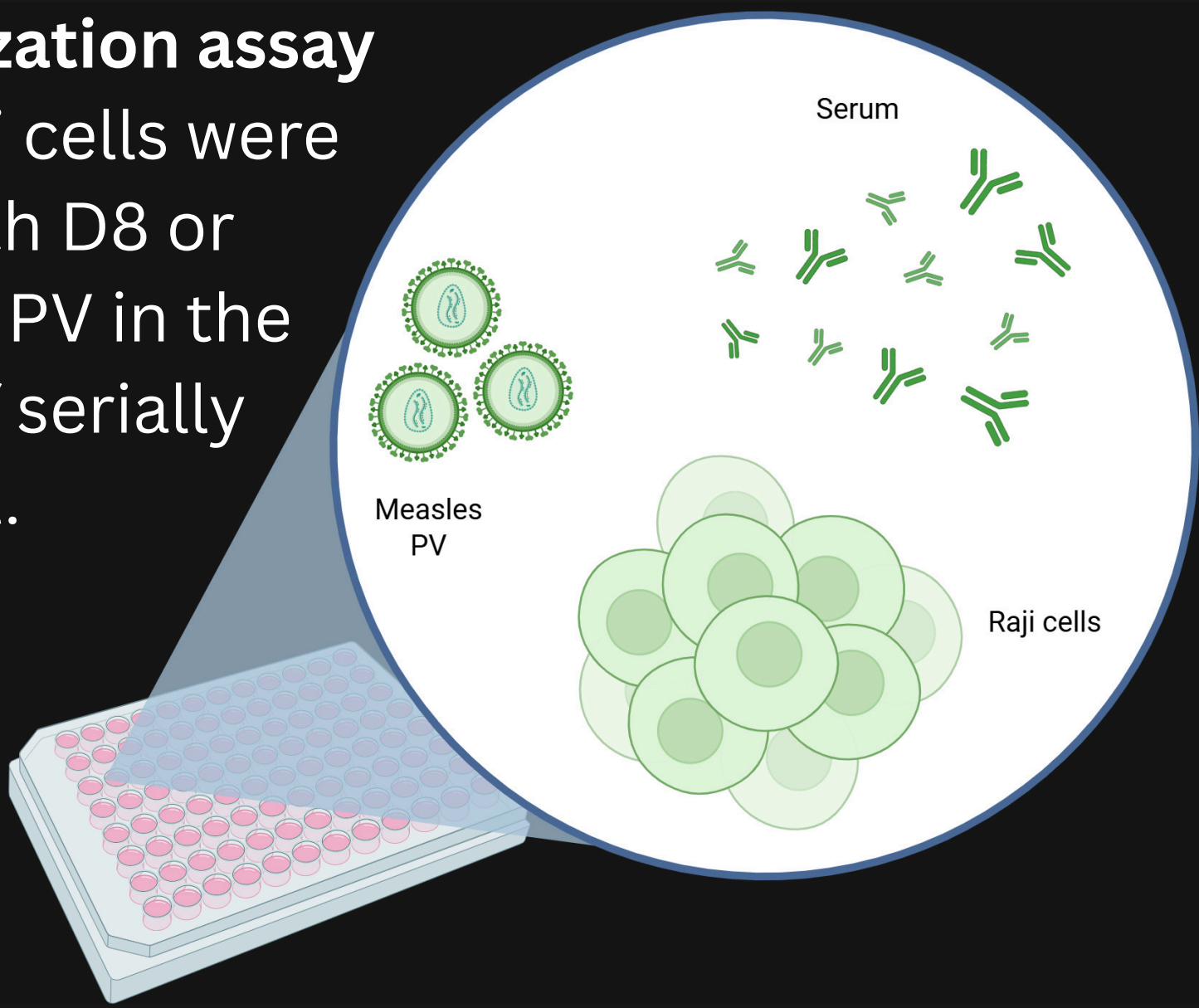
## METHODS

### PV production: Edmonston / D8



### PV neutralization assay (PVNA):

Raji cells were infected with D8 or Edmonston PV in the presence of serially diluted sera.



### Calculation:

% Neutralization, IC50 and nAb concentration (mIU/mL)

$$\%Neutralization = \frac{RLU_{max} - RLU_{experimental}}{RLU_{max} - RLU_{min}}$$

- IC50 was calculated for each sample and for the 3rd WHO Measles International Standard NIBSC 97/648 diluted at 120 mIU/mL (IS120): LL.4 fits on responses normalized to 0–100% of PV neutralization:

0 % - RLumin (Only cells, no serum, no PV)

100 % - RLUMax (Cells + PV, no serum)

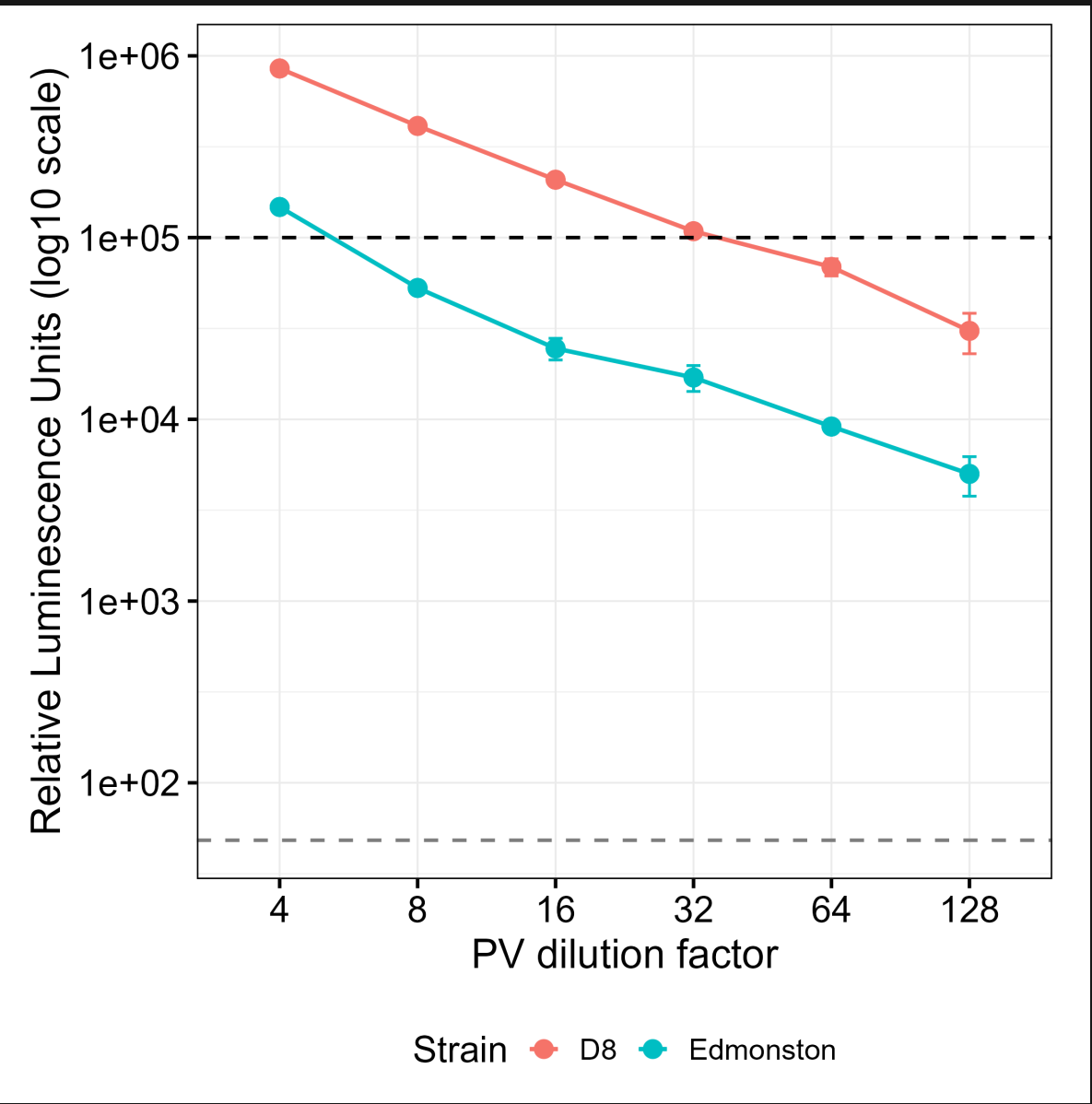
- IC50 values were calibrated to IS on the same plate and strain:

$$mIU/mL = 120 \times \frac{IC50 (sample)}{IC50 (IS120)}$$

- Higher mIU/mL = greater neutralizing activity.

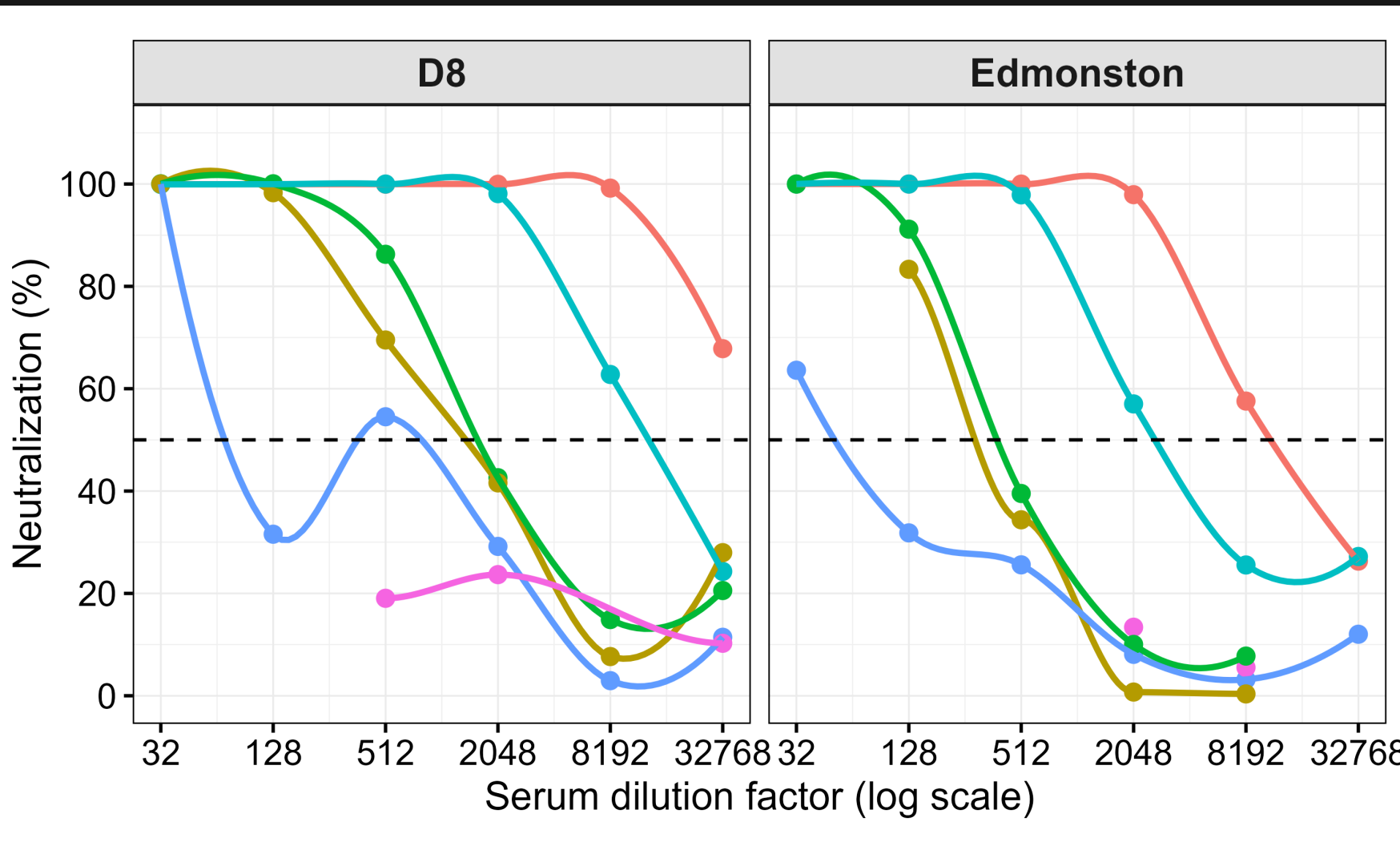
## RESULTS

### Titration of measles PV stocks on Raji cells.



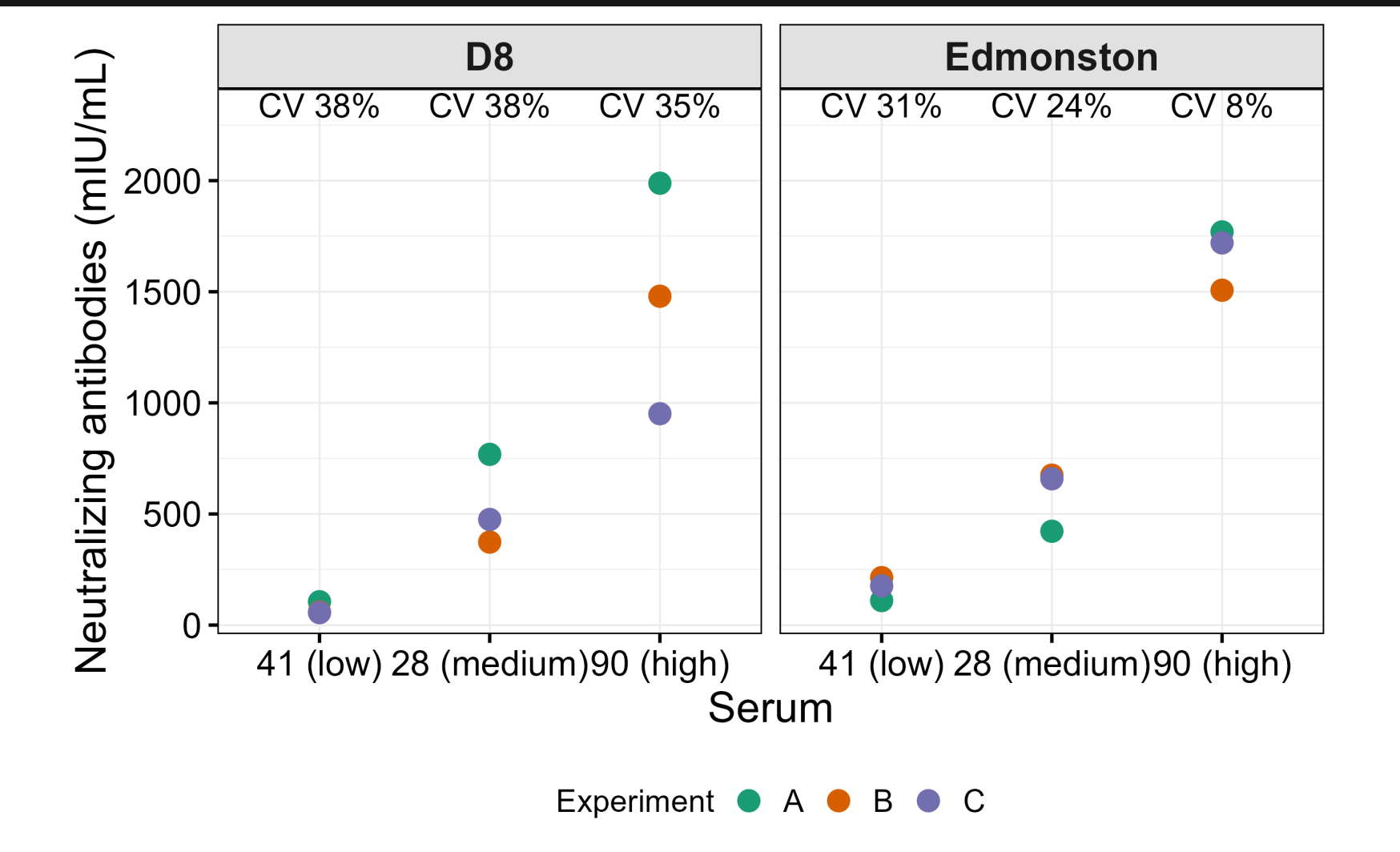
- PV titration: Serial dilutions (×2) of each PV + Raji cells.
- Grey dashed line: the cell-only background (no PV).
- Black dashed line: the target minimum input for neutralization assays (≥10<sup>5</sup> RLU at the chosen PV dilution).

### Typical MePV neutralization curves across serum dilutions.



- Points show measured % neutralization; lines are 4-parameter log-logistic fits (LL.4).
- Poorly neutralizing sera do not cross the dotted line- 50% of the luminescence value of the PV control (maximum RLU signal, no serum).
- Sera in the test range all cross the 50% neutralization and yield reportable titers (IC50).
- Highly neutralizing sera do not cross the 50% neutralization and must be retested at a higher initial dilution.

### Inter-assay repeatability of neutralizing antibodies (mIU/mL) for three sera classified by IgG level.



### Higher measles-specific IgG level corresponds with higher MePV neutralizing antibodies.

| Sample  | IgG level (mIU/mL) |        | PVNA (mIU/mL) |           |
|---------|--------------------|--------|---------------|-----------|
|         | Euro Immun         | Serion | D8            | Edmonston |
| 41 low  | 132                | 191    | 73            | 166       |
| 28 med  | 405                | 558    | 539           | 585       |
| 90 high | >1500              | 2782   | 1473          | 1665      |

## CONCLUSION:

- Measles pseudovirus neutralization, calibrated to the 3rd WHO Measles International Standard, provides comparable neutralizing antibody titers (mIU/mL) readouts for vaccine (Edmonston) and circulating (D8) strains.
- The assay discriminates low/medium/high IgG samples and shows acceptable inter-assay CVs, supporting its use for sero-monitoring and vaccine immunogenicity studies.

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